

Efficient Self-Supervised Grading of Prostate Cancer Pathology

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Abstract—Prostate cancer grading, using the International Society of Urological Pathology (ISUP) system, for treatment decisions is highly subjective and requires considerable expertise. Despite advances in computer-aided diagnosis systems, few have handled efficient ISUP grading on whole slide images (WSIs) of prostate biopsies based only on slide labels. In this scenario, TSOR is developed, where a novel task-specific self-supervised learning (SSL) framework is used for patch-level pretraining. This is fine-tuned using ordinal regression for WSI-level ISUP grading. One of the main challenges faced by deep learning (DL) in ISUP grading, is the learning of patch-level features based on slide labels. Though using models pretrained at patch-level using SSL or other paradigms is the most obvious choice here, the diversity of training samples plays a crucial role in effective pretraining. However, pretraining on a large database of different histopathology images becomes computationally expensive. Therefore, a patch-level dataset (relatively balanced with respect to the patch-level grades) is initially created for effective SSL-based pretraining. As stain-variation across centers leads to difficulty in generalization, an additional loss term is incorporated to effectively learn the stain-agnostic patch-level features. As it is desirable that misclassification be as close as possible to the actual grade, in medical images, we fine-tune the pretrained network for WSI-level ISUP grading using an ordinal regression-based approach. Experimental results on the most extensive prostate cancer grade assessment (PANDA) challenge dataset, and the SICAPv2 dataset, demonstrate the effectiveness of TSOR compared to state-of-the-art (SOTA) methods.

Index Terms—Histopathology, International Society of Urological Pathology (ISUP) grading, prostate cancer, self-supervised learning (SSL), whole slide images (WSIs).

I. INTRODUCTION

PROSTATE cancer is the second-most common cancer that affects men worldwide [1]. The gold standard for diagnosing and grading it, for determining the prognosis, typically involves Gleason grading of biopsies. A general workflow of pathologists for grading prostate biopsies includes the collection of a small amount of prostate tissue, laminating and staining with hematoxylin and eosin (H&E), followed by its staging. The Gleason grading system, with an inverse

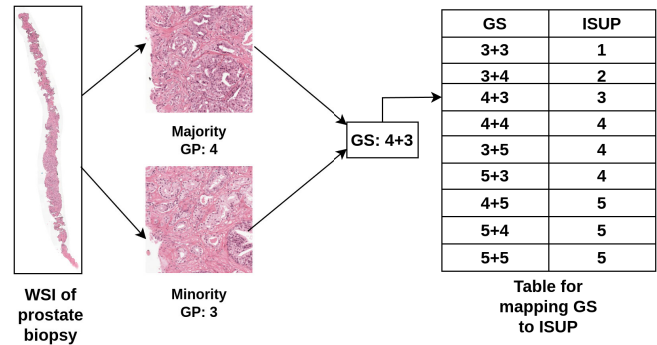


Fig. 1. ISUP grading system, based on GP and corresponding Gleason score (GS).

correlation between the degree of tissue gland differentiation and the Gleason grade (GG), contains three values ranging from 3 to 5 [2]. The new International Society of Urological Pathology (ISUP) grading system [3], as shown in Fig. 1, considers the first and second-most frequent Gleason pattern (GP), in terms of proportion and severity within a prostate biopsy, followed by mapping it to an ISUP grade (1)–(5). Benign biopsies are assigned an ISUP grade of 0.

The entire procedure for grading prostate biopsies by pathologists is cumbersome, requires considerable expertise, and is highly prone to intraobserver and interobserver variability. This highlights the need for computer-assisted prostate cancer diagnosis/progression systems [4]. With the advent of digitized biopsies or whole slide images (WSIs), the increased availability of graphic processing units (GPUs) at reasonable cost, and large publicly available datasets, the development of deep learning (DL)-based automated systems in histopathology has increased manifold [5], [6]. DL [7], [8], particularly convolutional neural networks (CNNs), work directly on the pixel values of an input image. Hence, it overcomes manual errors caused by inaccurate segmentation and/or hand-crafted feature extraction of shallow learning.

A. Related Work: Limitations and Motivation

The WSIs contain billions of pixels, and so it becomes computationally infeasible for DL models to process them directly. Thus, each WSI is represented as a collection of smaller patches, and a WSI-level prediction is obtained based on the patches. Generally, for cancer grading from WSIs, a DL model needs to learn a particular pattern corresponding to each grade. This feature learning is best possible with

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supervised training at the patch level [2], [3]. However, to identify the correct ISUP grade, the feature learning should be more robust to identify both the primary and secondary GPs from WSIs. Several algorithms [2], [9], [10], [11], [12] for ISUP grading initially trained patch-level supervised classifiers to predict GPs. The features or grade distribution derived from the patches were then aggregated by different pooling techniques or fed into machine-learning classifiers like k -nearest-neighbors (kNNs), multilayer perceptron, and random forest to determine the corresponding WSI grade.

Given the large size of digitized biopsies, it is often impossible to obtain expensive patch-level annotations; thereby, making it difficult to train such fully supervised DL models at patch level. In the absence of patch-level supervision, the multiple instance learning (MIL)-based weakly supervised algorithms can be used to train the DL models. Weakly supervised learning, including MIL [13], predicts the grade of a WSI based on slide-level labels (such as the ISUP grade) and a WSI-level representation derived from the patches. The use of MIL in histopathology literature includes instance- and bag-based approaches [5]. Instance-based MIL classifiers are trained on patch labels inferred from WSI labels, with MIL pooling for WSI-level prediction. However, bag-based MIL classifiers combine patch embeddings to obtain a WSI-level representation for training with WSI labels. Due to its robustness, the bag-based MIL approaches were mostly used in the following state-of-the-art (SOTA) algorithms.

Several bag-based MIL approaches focused on refining feature aggregation through multiscale patch embeddings in DSMIL [14], instance-level clustering in CLAM [15], and cross-slide contrastive learning in SCL-WC [16]. In addition, many frameworks aimed to capture spatial and contextual relationships among patches using transformer-based architectures, such as TransMIL [17], HIPT [18], and CEMIL [19], or through a Mamba-based architecture in 2DMambaMIL [20].

Apart from these generic MIL frameworks, different variants of the bag-based MIL approach were also explored specifically for ISUP grading in [1], [21], and [22]. The mean-squared error loss was used in conjunction with the bag-based MIL approach in [21]. The extracted features for ISUP grading were refined by incorporating multichannel and multispatial attention in a CNN [1]. The ISMIL framework [22], on the other hand, improved the recognition of small tumor regions by intensively sampling crucial regions. Further, to capture contextual information among the patches, Behzadi et al. [23] represented each WSI as a graph and employed a graph convolutional network (GCN) to grade each of them. Further, several recent studies have explored prostate cancer grading from different perspectives, including federated learning [24], interpretability [25], and multimodal data integration [26].

Among these histopathology-based algorithms, the generic MIL frameworks in [15] and [17] and ISUP grading-specific frameworks in [1] and [21] used ImageNet-pretrained models for patch-level feature extraction. However, these models were suboptimal for histopathology due to their bias toward natural image features. Due to this, several generic MIL frameworks [14], [16], [19], [20] and ISUP grading-specific frameworks [12], [22], [23] considered self-supervised learn-

ing (SSL)-based models, pretrained on relatively easy-to-obtain unlabeled histopathology data, for patch-level feature extraction. In addition, recent research has also experimented with SSL-based frameworks for prostate cancer detection based on other imaging modalities such as magnetic resonance imaging [27] and ultrasound [28].

SSL aims to reduce the manual annotation burden by learning effective feature representations from a set of unlabeled data [29]. This process of feature learning, through some auxiliary tasks, is termed pretraining. It can be task-specific or task-agnostic, depending on whether or not the training samples and the auxiliary task are considered to be solely based on the target task. The pretrained model is then fine-tuned for the target task based on a limited amount of labeled data. Due to the effectiveness of the SSL paradigm [29], [30], the development of histopathology-specific foundation models has gained momentum in recent times [31], [32], [33], [34]. These involve DL architectures, pretrained using SSL or other paradigms on large-scale histopathology datasets, to learn rich, domain-specific visual representations. The primary motivation is to learn robust, domain-specific features to handle variability in different histopathological images; thus reducing the dependence on large, annotated, task-specific datasets.

However, even these MIL-based approaches, based on task-agnostic SSL, makes it challenging to learn fine-grained, discriminative features corresponding to individual GPs. In MIL, the WSI-level representation combines features from multiple heterogeneous patterns (e.g., primary and secondary GPs), and the supervision signal is only available at the WSI-level (not at the patch-level). In many cases, the secondary GP is present in only very small amount relative to the primary GP, or is found to overlap with the primary GP in the same patches. In such situation, unless the GP discriminatory power of the model is very robust, only the dominant primary GP would be recognized among the aggregated patch-level features in MIL; leading to misclassification. For example, if the GP 5 could not be identified in a sample having ISUP grade 5 ($4 + 5$), then it could be misclassified as grade 4 ($4 + 4$). As observed from Fig. 1, to differentiate between ISUP grades 2 ($3 + 4$) and 3 ($4 + 3$), where the GPs differ only in proportion, the patch-level features should be very precise. Not only is it computationally expensive to pretrain such task-agnostic frameworks and foundation models using a large database of high-resolution histopathological images of different types of cancer, but their learning is also effective only for capturing general characteristics of the histopathology images. Cancer patterns are typically heterogeneous and unique in prostate cancer images. Hence, it becomes desirable to use a task-specific SSL framework with prostate cancer-specific patches to learn the fine-grained features.

However, randomly sampling unlabeled patches from prostate WSIs results in large datasets with huge class imbalance. Precisely, the number of benign patches becomes maximum, with GG 5 patches being minimum [2]. In this scenario, SSL-based models will not only take huge training time but face difficulty in learning discriminating features from the corresponding GP [5], [29]. In order to circumvent this

problem, we create a relatively balanced patch-level dataset with respect to GG by pseudo-labeling the patches with MIL. We adopt a novel hybrid MIL approach to train our model for the pseudo-labeling of patches. It involves simultaneous training of instance-based and bag-based classifiers, to balance the tradeoff between the two.

Different practical problems in analyzing histopathological images collected from different centers include tissue preparation, scanner variation, image resolution, and staining variation. Among them, the scanner-induced visual changes can mostly be addressed through standard color/intensity augmentations and other differences can also be handled by standard preprocessing pipelines in MIL. In contrast, stain variation produces large, systematic shifts in color distribution that cannot be properly addressed by generic augmentations [35], [36]. Hence, we incorporate a novel module during the task-specific SSL-based pretraining that enforces consistency between two stain-augmented views of the same patch. This helps in learning the stain-agnostic morphological characteristics of the GP, thereby improving cross-center generalization.

Error in medical image grading should be as close as possible to the actual grade for a model to be called efficient. When there exists an ordering between classes, mapping of the objective function as an ordinal regression [37], [38] allows a neural network to minimize misclassification through ranking. The ISUP grading is a multiclass classification problem, with inherent ordering between the grades. In contrast, traditional cross-entropy loss-based classification in the existing methods assumes that the error of misclassifying ISUP grade 1 as 4 is just as bad as misclassifying it as 2. Our model minimizes the distance between the predicted and actual labels through ordinal regression to fine-tune the pretrained network. Thus, even when the features are not precise enough to identify the correct ISUP grade, our novel fine-tuning approach still tries to ensure that the predictions are at least closer to the actual grade.

B. Contribution

A novel and efficient SSL-based ISUP grading system TSOR is developed for prostate WSI, based on slide-level labels. A task-specific SSL framework, trained on a well-curated dataset, helps to effectively learn the stain-agnostic patch-level features of GPs. This is unlike the previous approaches where patch-level annotations were needed to learn the (patch-level) features. Neither is it as computationally expensive as the other pretraining frameworks, typically employed to learn general patch-level features. The novel pretraining in TSOR also tries to address the problem of generalizability due to stain variation across different datasets. Taking into account the ordinal nature of the classes, it is then fine-tuned for the ISUP grading task using a novel ordinal regression-based approach. This is in contrast to the traditional cross-entropy loss-based classification, where the ordinal nature of the classes is not considered. Thus, our approach ensures an erroneous grade to be as close as possible to the actual one. The block diagram in Fig. 2 outlines the interconnections between the individual modules involved, in

the TSOR framework, with the main contributions summarized below.

- 1) Preparation of patch-level dataset, relatively balanced with respect to GGs, for effective self-supervised pre-training. This is achieved through the labeling of the patches in a novel hybrid MIL framework.
- 2) Design of a novel task-specific SSL framework with a new loss term based on stain augmentation. This allows the model to learn the stain-agnostic morphological features of the GPs.
- 3) Fine-tuning of the pretrained network, in terms of ordinal regression, for ISUP grading.

The effectiveness of the model, compared to the SOTA-related methods, is established through validation on prostate cancer grade assessment (PANDA) challenge [39] and SICAPv2 benchmark datasets [2].

The rest of the article is organized as follows. Section II introduces the TSOR, a new task-specific SSL framework fine-tuned using ordinal regression, for ISUP grading. Section III describes the experiments carried out to evaluate the efficacy of TSOR with respect to related SOTA algorithms, along with suitable quantitative and qualitative results. Finally, Section IV concludes the article.

II. METHODOLOGY

Let $\{s_n\}_{n=1}^N$ be a set of N WSI of prostate biopsies. Each $s_n = \{x_{n,i}\}_{i=1}^l$ is represented as a bag of l patches, where $x_{n,i}$ denotes the i th patch cropped from the n th slide. The aim is to develop a model to predict the ISUP grade of a WSI using only slide-level information, as available from clinical records in terms of GG. This corresponds to the primary GG $z_{n,pg}$, secondary GG $z_{n,sg}$, and ISUP grade $z_{n,isup}$ of each s_n . Our TSOR framework achieves this goal by using three distinct modules, as described below.

A. Module 1: Preparation of Training Data for SSL

A patch-level dataset is created for an effective SSL, such that it is relatively balanced with respect to GGs. A DL model, based on the hybrid MIL framework, is trained to label patches using slide-level GGs in an end-to-end fashion. The model training procedure is described in Fig. 3.

The rationale behind considering the hybrid MIL framework is to balance the tradeoff between the bag-based and instance-based MIL approaches. The workflow of instance-based MIL involves supervised training of a patch-level classifier, with aggregation of patch-level outputs to generate a WSI-level prediction. Since patch-level annotations are not known, these approaches typically label all patches from a cancerous WSI as cancerous. As this results in noisy training of the patch-level classifier, the patch labels cannot be predicted with high confidence using such frameworks. Moreover, each cancerous WSI can have multiple GPs. The bag-based MIL, on the other hand, entails extracting and combining the features of all patches to generate the WSI-level features. These are used in the supervised training of the bag-level classifier with the WSI labels. Although there is no noise involved during this

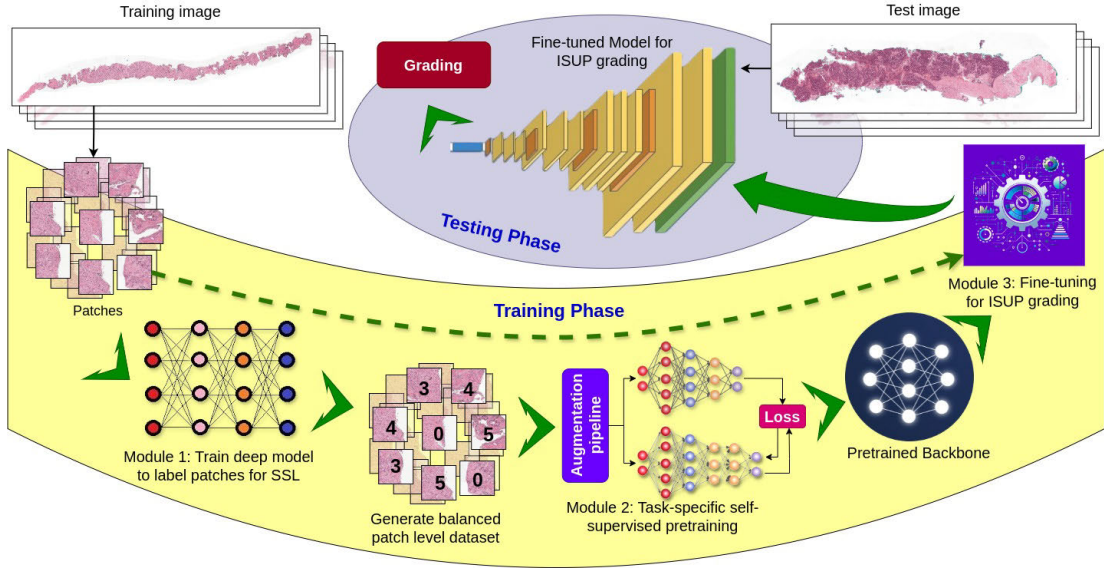


Fig. 2. Schematic representation of the proposed framework TSOR. In Module 1, a DL model is trained using WSI-level labels to generate a patch-level dataset, relatively balanced with respect to the GGs. This dataset is then used for SSL-based pretraining in Module 2, enabling the model to learn effective patch-level features. The pretrained model is then fine-tuned for ISUP grading in Module 3.

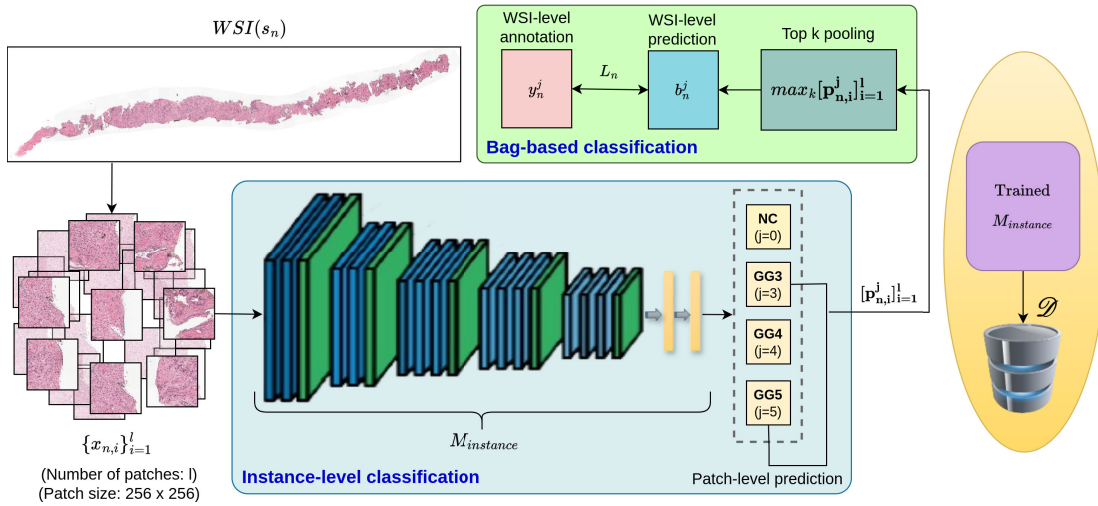


Fig. 3. Module 1 consists of the hybrid MIL framework, trained to label the patches and create patch-level dataset for SSL. Each WSI is treated as a bag of l patches. In the first part of the framework, M_{instance} predicts each patch's probability of being benign, GG3, GG4, and GG5. In the second part, the patch-level probabilities in each WSI are converted to the corresponding WSI-level probabilities for each GG by top k pooling. The trained M_{instance} is then used to label the patches and generate a patch-level dataset $\mathcal{D}$.

training, the models remain unsuitable for the prediction of patch labels [5].

The first part of the proposed hybrid MIL framework (in TSOR) consists of an instance-level classifier M_{instance} that predicts the probability $p_{n,i}^j$ of a patch $x_{n,i}$ to belong to one of the four classes j . Here, $j = 0$ indicates benign, with j ranging from 3 to 5 for GG3, GG4, GG5. The second part involves bag-based classification, where all patch-level probabilities $p_{n,i}^j$ in each WSI s_n are converted to the corresponding bag-level probabilities $[b_n^j]$ for GG $j = 3, 4, 5$. We have

$$[b_n^j] = \left\lceil \frac{\sum \max_k [p_{n,i}^j]_{i=1}^l}{k} \right\rceil \quad (1)$$

where l denotes the total number of patches considered from each WSI, k denotes the number of top-probability patches considered for computing the WSI-level probability, and each component b_n^j represents the average of the corresponding top k (selected by $\max_k$) $p_{n,i}^j$ of the WSI. If a WSI has a small amount of GP, then averaging all $p_{n,i}^j$ s can dilute the signal and reduce the WSI-level probability b_n^j . This might lead to inaccurate predictions. Using a maximum of $p_{n,i}^j$ s is susceptible to single-patch noise and ensures only a correct classification of patches possessing a large amount of GP. In this context, averaging the top $k p_{n,i}^j$ increases the robustness of the model to single-patch noise, while also correctly classifying the patches containing moderate amounts of GP. The sensitivity analysis for the parameter k is provided in Section III-C.

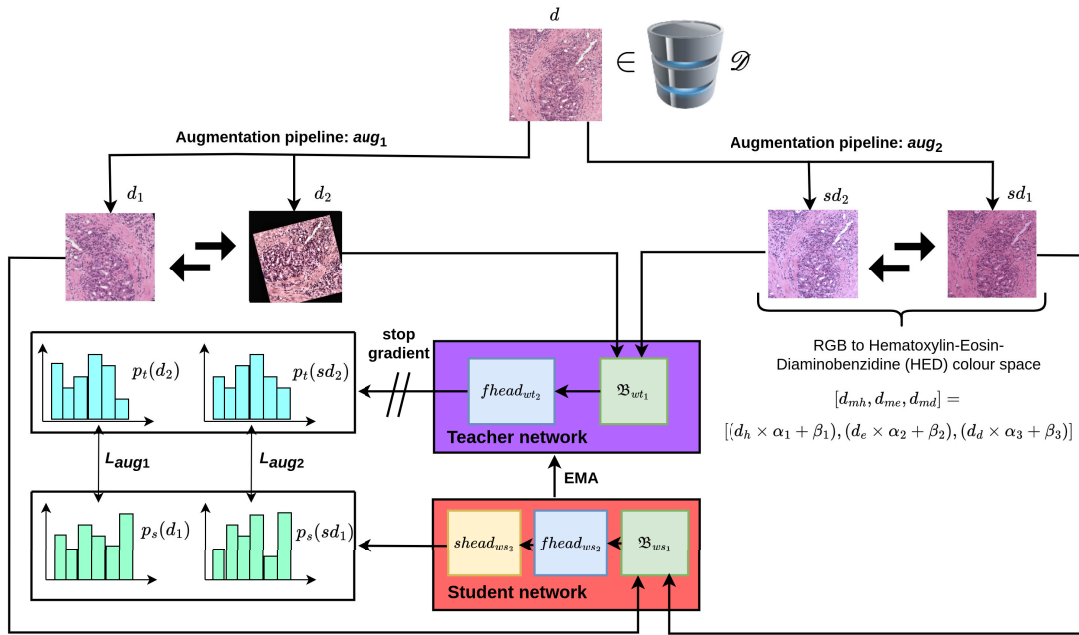


Fig. 4. Module 2 of TSOR represents the task-specific self-supervised pretraining procedure. Among the two augmentation pipelines, aug_1 generates two augmented views d_1 and d_2 based on general augmentations and aug_2 generates two views sd_1 and sd_2 based on stain augmentation. The pairs (d_1, sd_1) and (d_2, sd_2) are then alternately passed into the student and teacher networks to make the loss symmetric. While L_{aug1} enforces consistency between the predictions for d_1 and d_2 , L_{aug2} does the same for sd_1 and sd_2 . The student network is then updated by the summation of these two losses and the teacher network is updated as EMA of the student's parameters.

Given that there exist two GGs corresponding to each WSI, we pose this as a multilabel binary classification problem with the loss defined as

$$L_n = \text{BCE}([b_n^j], [y_n^j]) \quad (2)$$

where BCE denotes the multilabel binary cross-entropy loss between predicted bag-based probability at WSI-level [see (1)] and the one-hot encoded target output $[y_n^j]$. In multilabel BCE, the BCE loss is computed independently for each GG and averaged across labels. The value of each y_n^j is 0 if the WSI does not exhibit GG j (in $z_{n,pg}$, $z_{n,sg}$), and 1 otherwise.

The trained deep model M_{instance} is then used to label the patches. An approximately equal number of patches are selected from each class to create the patch-level dataset $\mathcal{D}$ for self-supervised pretraining. Patches belonging to each class are initially sorted, based on tissue percentage, followed by the selection of the top patches having greater tissue content.

B. Module 2: Task-Specific and Stain-Agnostic SSL Pretraining

Next, a deep model is designed based on a task-specific and stain-agnostic SSL framework. It follows the teacher-student knowledge distillation approach [40]. The aim is to effectively learn the stain-invariant morphological features of GPs using a novel loss function. The training procedure is summarized in Fig. 4. There are two augmentation pipelines involved. The aug_1 pipeline involves common strategies such as random color jitter, grayscale conversion, Gaussian blur, and horizontal flip. Given an image patch $d \in \mathcal{D}$, the augmented views d_1 and d_2 are generated from aug_1 .

In order to learn the stain-invariant characteristics for patch d , two different stain-augmented views sd_1 and sd_2 are generated using the aug_2 pipeline. It simulates a range of realistic H&E-stained images. First, each patch sample d is transformed from RGB to hematoxylin-eosin-diaminobenzidine (HED) color space and decomposed into three channels d_h , d_e , and d_d based on color deconvolution [41]. Each of the HED channels is modified as

$$[d_{mh}, d_{me}, d_{md}] = [(d_h \times \alpha_1 + \beta_1), (d_e \times \alpha_2 + \beta_2), (d_d \times \alpha_3 + \beta_3)], \quad (3)$$

with random factor $\{\alpha_i\}_{i=1}^3$, bias $\{\beta_i\}_{i=1}^3$, being sampled from bounded uniform distributions centered around unity and zero, respectively, that is, $\alpha \in [1 - as, 1 + as]$ and $\beta \in [-as, as]$, with $as \in [0, 1]$ controlling the magnitude of the stain perturbation. Smaller values of $as \in [0.05, 0.2]$ were adopted, as they introduce moderate, morphology-preserving stain variations that are well-aligned with the objective of consistency-based SSL. Such perturbations encourage the network to learn stain-invariant yet task-relevant features, while larger values can lead to unrealistic color distortions and reduced training stability [29], [35], [36].

The modified channels are then combined, with the sample reconverted to RGB space, to generate a stain-augmented image sd_1 . The same procedure is repeated by sampling different values $\{\alpha_i\}_{i=1}^3$, $\{\beta_i\}_{i=1}^3$, to obtain the stain-augmented image sd_2 .

The backbone of both teacher and student networks has a similar architecture, with the teacher backbone represented as $\mathcal{B}_{wt1}$ and the student backbone denoted by $\mathcal{B}_{ws1}$. Here, $wt1$ and $ws1$ denote the corresponding model weights. The heads of both networks are asymmetric, to avoid model

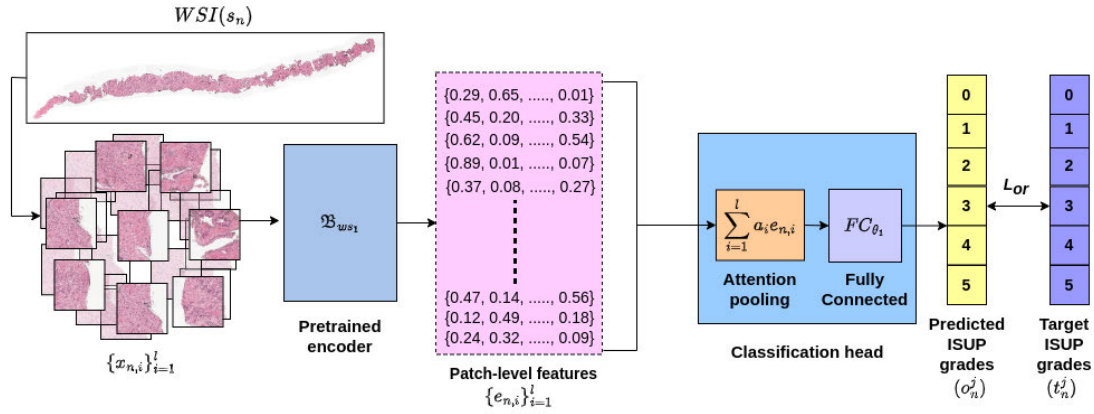


Fig. 5. Module 3 of TSOR represents the fine-tuning procedure for ISUP grading. Each ISUP grade is mapped to a vector such that the difference between classes follows an ordinal scale. A custom classification head, consisting of the attention and fully connected layers, is added on top of the backbone of the pretrained student network. While the attention pooling layer weighs each patch's features by its contribution in the WSI-level prediction, the fully connected layer predicts the final ISUP grade based on it.

degeneration [5]. Specifically, the head of the student model comprises a pair of MLPs as fhead_{ws_2} and fhead_{ws_3} . The head of the teacher model consists of only one MLP as fhead_{wt_2} .

The prediction of the student network, for any patch img , is obtained as

$$p_s(\text{img}) = \text{fhead}_{ws_3}(\text{fhead}_{ws_2}(\mathfrak{B}_{ws_1}(\text{img}))). \quad (4)$$

Analogously, the prediction of the teacher network corresponding to any patch img is represented as

$$p_t(\text{img}) = \text{fhead}_{wt_2}(\mathfrak{B}_{wt_1}(\text{img})). \quad (5)$$

The parameters of the student network are updated by minimizing the new loss function

$$L_{\text{total}} = L_{\text{aug}_1} + \lambda L_{\text{aug}_2}, \quad (6)$$

where λ is a hyperparameter used to adjust the contribution of L_{aug_2} . Here, λ was chosen in the range $[0.01, 0.03]$ after several experiments. Since gradients do not flowthrough the teacher network during training, the parameters are updated through the exponential moving average (EMA) from the student's parameters.

Note that the loss component L_{aug_1} , based on the aug_1 pipeline, is expressed as

$$L_{\text{aug}_1} = \{\text{mse}(\text{L2norm}(p_s(d_1)), \text{L2norm}(p_t(d_2))) + \text{mse}(\text{L2norm}(p_s(d_2)), \text{L2norm}(p_t(d_1)))\}/2. \quad (7)$$

Initially, d_1 is passed to the student network and d_2 is transmitted to the teacher network. The corresponding mean-squared error (mse) between the L2-normalized prediction $p_s(d_1)$ of the student network [see (4)] and the target output $p_t(d_2)$ of the teacher network [see (5)] is computed. To make L_{aug_1} symmetric, the same procedure is repeated by passing d_1 to the teacher network and d_2 to the student network. The corresponding predictions $p_s(d_2)$ and $p_t(d_1)$ are now calculated from (4) and (5).

Analogously, sd_1 (sd_2) are generated from the aug_2 pipeline and passed to the student (teacher) networks, respectively, and vice versa, as before. The corresponding loss component L_{aug_2}

is evaluated as in (7), but with sd_1 and sd_2 instead of d_1 and d_2 , respectively.

C. Module 3: Fine-Tuning for ISUP Grading

The backbone of the pretrained student network is adapted in a novel manner, for the task of ISUP grading, by posing it as an ordinal regression problem. The training framework is depicted in Fig. 5. Each label or ISUP grade $z_{n,\text{isup}}$ is converted to the corresponding target vector $[t_n^j]_{j=1}^5$, where the sum of the five elements is equal to the corresponding ISUP grade. For example, ISUP grade 0 is mapped to $[0, 0, 0, 0, 0]$ and ISUP grade 2 is mapped to $[1, 1, 0, 0, 0]$. This kind of label encoding, as opposed to the one-hot encoding of traditional classification models, ensures that the difference between classes follows an ordinal scale. For example, the difference between ISUP grades 2 and 5 is greater than that between grades 2 and 3 when such encoding is used. It helps to ascertain that errors made by the deep model are close to the actual grade. This is highly desirable in grading medical images.

The backbone of the pretrained student network $\mathfrak{B}_{ws_1}$ uses a custom classification head during training for ISUP grading. It comprises an efficient attention-based MIL pooling operator for improved interpretability, followed by a fully connected layer. The convolutional part of $\mathfrak{B}_{ws_1}$ generates low-dimensional patch-level embedding

$$\{e_{n,i}\}_{i=1}^l = \mathfrak{B}_{ws_1}(\{x_{n,i}\}_{i=1}^l), \quad (8)$$

with l denoting the number of patches from each WSI. Patches are concatenated using the attention-based MIL pooling operator, evaluated as

$$a_i = \frac{\exp\{W_{\theta_2}^T \tanh(V_{\theta_2}(e_{n,i}))\}}{\sum_{j=1}^l \exp\{W_{\theta_2}^T \tanh(V_{\theta_2}(e_{n,j}))\}} \quad (9)$$

where V_{θ_2} and W_{θ_2} are two fully connected layers, $\tanh$ is the activation function, T denotes transpose operation, and $\exp$ is the exponentiation operation. Here, a_i provides each embedding $e_{n,i}$ with the context about its role in contributing

TABLE I
DISTRIBUTION OF PROSTATE BIOPSY WSI USED IN LEARNING

ISUP	Training (PANDA)	Validation (PANDA)	Testing (PANDA)	Testing (SICAP)
0	922	166	272	36
1	922	166	271	14
2	921	166	272	22
3	922	166	273	23
4	904	116	255	18
5	913	100	253	42

to the WSI label. Consequently, the fully connected layer FC_{θ_1} predicts the output vector

$$[\mathbf{o}_n^j]_{j=1}^5 = FC_{\theta_1} \left(\sum_{i=1}^l a_i, e_{n,i} \right), \quad (10)$$

with the training loss being computed as

$$L_{or} = \text{BCE}([\mathbf{o}_n^j]_{j=1}^5, [\mathbf{t}_n^j]_{j=1}^5). \quad (11)$$

III. IMPLEMENTATION AND EXPERIMENTAL DETAILS

This section describes the publicly available prostate WSI datasets used along with implementation details of the algorithms. This is followed by experimental results obtained for the detection and grading of prostate cancer. It includes ablation studies, as well as comparison with related SOTA methods.

A. Datasets Used

The model was trained with the largest multicenter prostate WSI dataset from the Kaggle PANDA challenge [42]. The testing phase was carried out on data originating from two different centers, that is, the PANDA and SICAPv2 datasets [2]. PANDA consists of 10 616 H&E-stained digitized biopsies, whose primary and secondary GG were labeled by expert pathologists. Considering the tradeoff between the magnification details and the associated computational complexity, we considered the intermediate-resolution ($5\times$) images for our experiments. The SICAP database, used for testing, consists of 155 WSI with global primary and secondary GG annotated by expert pathologists. As the images in SICAP are available at $10\times$ magnification, they were downsampled to $5\times$ resolution for uniformity.

The distribution of WSIs, used for training, validation, and testing with respect to ISUP grades, is provided in Table I. ISUP 0 images are not required in Module 1. Around 4000 patches of each class, from the training partition of WSIs, are used for SSL pretraining of Module 2. Finally, Module 3 works on the data splits summarized in the table. Annotations of all GP are available in case of SICAP, whereas only one center's annotations (covering 49% of the data) are provided for PANDA.

B. Implementation Details

The lightweight EfficientNet-B1 [43] was used as the backbone for all deep models used here, with patches of size 256×256 as input. Patches from each slide were first sorted,

based on average pixel intensity, in ascending order. The top 36 patches, having the lowest average pixel intensity, were considered from each slide for training in Modules 1 and 3. Based on prior studies [1], [22], [23], [39], this intensity-based patch filtering is used solely to exclude background-dominated patches and ensure sufficient tissue content per WSI. The choice of 36 patches per WSI was empirically determined to provide a good trade-off between tissue coverage and computational efficiency, enabling stable training with a reasonable WSI batch size. Runtime augmentation was used at patch level during training in Modules 1 and 3, using affine transforms such as random flipping, rotation, and scaling, along with color augmentations such as random changes in brightness, contrast, saturation, and hue. Adam optimizer was used in all three modules, along with the cosine annealing learning rate scheduler, involving an initial learning rate of $3e^{-4}$. Considering batch sizes of 8, 40, and 8, in Modules 1, 2, and 3, respectively, the corresponding deep models were trained for 35, 50, and 30 epochs, respectively. The implementations were made in the Pytorch framework (version 1.13.1) using Python (version 3.9) with a dedicated GPU (NVIDIA RTX A6000 of capacity 48 GB).

C. Results: Ablations and Comparisons

The metrics used to evaluate the different modules of TSOR include the $F1$ score, the area under the receiver operating characteristic (ROC) curve (AUC), accuracy, and Cohen's quadratic kappa [10]. The Cohen quadratic kappa is a statistical metric used to measure the agreement between two raters, while accounting for the possibility of agreement occurring by chance. It is a weighted version of Cohen's Kappa, where disagreements get penalized more when they are larger. It is expressed as

$$\text{Kappa} = 1 - \frac{\sum_{i,j} W_{ij} O_{ij}}{\sum_{i,j} W_{ij} E_{ij}} \quad (12)$$

where O_{ij} is the observed confusion matrix, E_{ij} is calculated as the outer product between the actual histogram vector of outcomes and the predicted histogram vector, and W_{ij} is the quadratic weight calculated as $W_{ij} = \left(\frac{i-j}{k-1}\right)^2$. Here, i and j are categories, and k is the number of possible categories.

The bar plot with error bars (representing mean $\pm$ std) in Fig. 6 depicts the performance of the hybrid MIL framework in Module 1 with respect to the $F1$ score. To evaluate the effect of the number of top-scoring patches (k) on model performance, we conducted a sensitivity analysis with $k = 1$, $k \in [3, 7]$, $k \in [18, 22]$, $k = 36$, considering a total of 36 patches per WSI (corresponding to approximately 3%, 14%–19%, 50%–61%, and 100% of the total patches, respectively). The model achieves its highest and most stable performance for $k \in [3, 7]$, showing both strong mean $F1$ scores and low standard deviations across GGs. This range effectively balances between focusing on the most discriminative regions and maintaining robustness against noise, which is critical for reliable patch-level classification later. Using only the top patch ($k = 1$) makes the framework highly sensitive to single-patch noise. This setting tends to favor only patches

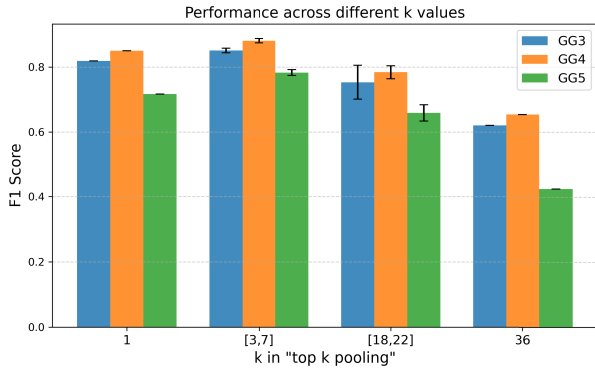


Fig. 6. Bar plot with error bars (mean $\pm$ std) showing sensitivity analysis for the “top k pooling” parameter k in Module 1.

containing large and distinct areas of a particular GP, leading to inconsistent generalization. For moderate k values ([18, 22]), performance declines considerably, accompanied by larger standard deviations. This indicates instability and potential inclusion of less relevant or confusing patches that dilute the class-discriminative signal. On the other hand, averaging over all patches ($k = 36$) results in a severe drop in performance, as the inclusion of background or nondiagnostic regions weakens the discriminative signal and reduces class separability. This ultimately impairs the patch-level classifier’s ability to distinguish between subtle histological patterns.

The deep model, for $k = 5$ and the most balanced $F1$ score across GGs, was used to label patches for the SSL pretraining step. Classwise $F1$ score is a widely adopted evaluation metric in multilabel learning [44]. In the MIL setting, these classwise $F1$ scores serve as a meaningful indicator of patch-level discrimination, as they reflect the model’s ability to correctly identify and aggregate discriminative patch evidence corresponding to each GP. Also, classwise $F1$ is particularly suitable in this context as it evaluates per-class discrimination quality independently, which is critical for generating diverse and informative pseudo-labels for SSL. Next, the network generated in Module 2, with the lowest validation loss, was used as the backbone for Module 3. The efficacy of the proposed framework TSOR is demonstrated for both prostate cancer detection and grading tasks. While detection aims to determine whether a prostate WSI is benign (ISUP 0) or malignant (ISUP 1–5), the grading task classifies the WSI into one of the six ISUP grades (0–5).

1) *Ablations*: The impact of the task-specific and stain-agnostic SSL pretraining, as well as ordinal regression in case of ISUP grading, was evaluated through extensive ablation studies for the detection and grading in test datasets. The best results are marked in bold.

a) *Detection*: Table II shows the ablation study for prostate cancer detection, over both test datasets, with respect to the performance metrics. Specifically, we evaluated Module 3 of our framework, pretrained using task-specific SSL [L_{aug_1} of (7)], task-specific and stain-agnostic SSL [L_{total} of (6)], and without SSL pretraining (ImageNet pretrained). A clear improvement can be observed when Module 3 is pretrained using task-specific and stain-agnostic SSL, as com-

TABLE II
PERFORMANCE OF MODULE 3 OF TSOR, ON THE TEST DATASETS, IN TERMS OF DIFFERENT METRICS

Data $\rightarrow$	SICAP			PANDA		
Method	Accuracy	F1 score	AUC	Accuracy	F1 score	AUC
Without SSL	0.8129	0.7600	0.9939	0.9167	0.9031	0.9675
Task-specific SSL	0.8968	0.8853	0.9988	0.9336	0.9342	0.9750
Task-specific & stain-agnostic SSL	0.9097	0.9013	1.0	0.9386	0.9399	0.9762

TABLE III
GRADING PERFORMANCE OF TSOR, ON THE TEST DATASETS, IN TERMS OF KAPPA

Dataset $\rightarrow$	SICAP	PANDA	Average
Without SSL & Without OR	0.7208	0.8250	0.7729
Without SSL & OR	0.8129	0.8542	0.8335
Task-specific SSL & OR	0.8612	0.8705	0.8658
Task-specific & stain-agnostic SSL & OR	0.8736	0.8843	0.8789

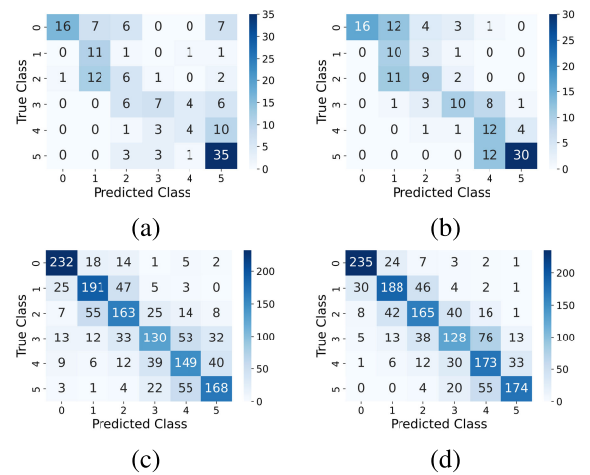


Fig. 7. Confusion matrices generated for ISUP grading, using Module 3 of TSOR: task-specific and stain-agnostic SSL with fine-tuning without OR on (a) SICAP, (c) PANDA; and task-specific and stain-agnostic SSL with fine-tuning and OR, on (b) SICAP, and (d) PANDA.

pared to the rest. The observation corroborates that our TSOR framework, pretrained using the same approach, performs consistently across both datasets; thereby, justifying its robustness.

A high AUC value across both datasets indicates that our TSOR is good at distinguishing between the positive and negative classes; thus achieving a good balance between true positives and false positives. This is highly desirable in an automated diagnostic system.

b) *Grading*: The ablation study for prostate cancer grading, by kappa score, is shown in Table III. We evaluated the impact of pretraining using the task-specific and stain-agnostic SSL framework, along with fine-tuning involving ordinal regression (OR), on the ISUP grading performance of Module 3. Here, fine-tuning “without OR” implies training using the more conventional one-hot encoding with cross-entropy loss. It is evident from the last column that our TSOR performed consistently well on test data. This highlights the robustness of TSOR to the domain shift caused by variations in stain color.

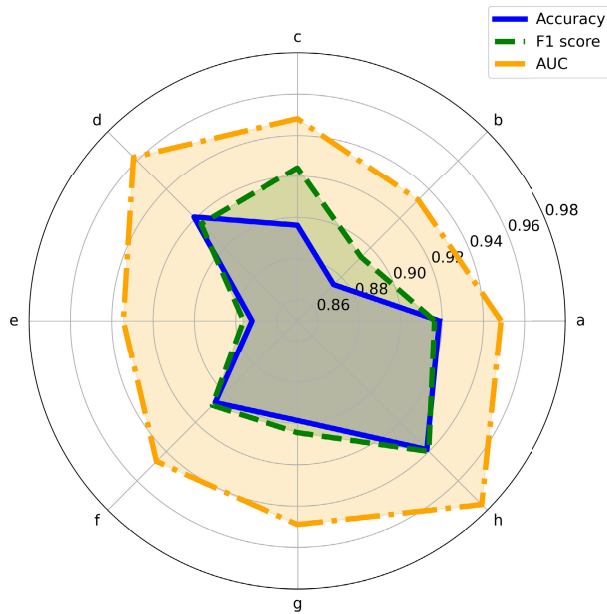


Fig. 8. Comparative performance of proposed (h) TSOR, in terms of Radar plot, with SOTA including (a) TMIL [17], (b) DSMIL [14], (c) CLAM [15], (d) SCL-WC [16], (e) HIPT [18], (f) CEMIL [19], and (g) 2DMambaMIL [20] for the detection of prostate cancer on PANDA test set.

The confusion matrices in Fig. 7 show that the prediction errors occur more in grades adjacent to the desired (true) one, when our OR-based approach (in lieu of the more conventional one-hot encoding scheme) is used for fine-tuning. Thus, in parts (b) and (d), the model avoids extreme misclassification in the top-right and bottom-left corners. This suggests improved class separation, leading to fewer severe prediction errors [compared to those in parts (a) and (c) of the figure]. For example, consider the incorrect predictions for true grade 5 in Fig. 7(a). These correspond to 3, 3, 1, for grades 2, 3, 4, respectively. Upon incorporating OR, the confusion shifts entirely to the adjacent predicted grade 4 in part (b). Besides, the number of predictions for grades 0, 1, 2 getting misclassified as grade 5 in part (a) becomes zero in part (b) of the figure. Comparing parts (c) and (d) of the figure demonstrates a reduction in the number of grade 3, 4, 5 images getting misclassified as grade 0 in part (d), along with a simultaneous increase in the number of mispredictions being made only in positions adjacent to the actual grade. It corroborates the effectiveness of employing ordinal regression, during fine-tuning, for ISUP grading.

2) *Comparative Study*: Here, we summarize the comparisons for the detection and ISUP grading of prostate cancer, with related SOTAs from the literature.

a) *Detection*: The performance of our TSOR was compared with CLAM [15], DSMIL [14], TMIL [17], SCL-WC [16], HIPT [18], CEMIL [19], and 2DMambaMIL [20], using accuracy, $F1$ score, and AUC metrics on the PANDA test set. The results are illustrated in the Radar plot in Fig. 8. It is observed that the TSOR model consistently outperformed the SOTA. Note that [15], [17] used ImageNet pretrained model to extract patch-level features. Li et al. [14], Wang et al. [16], Rahman et al. [19], and Zhang et al. [20] employed

TABLE IV
COMPARATIVE STUDY WITH OTHER SOTA METHODS FOR ISUP GRADING ON THE PANDA DATA

Method	Kappa score
Self-learning system (Features + Average + MLP) [10]	0.8245
Self-learning system (Features + Average + kNN) [10]	0.7927
GG% + kNN [9]	0.8152
GG% + MLP [2]	0.8229
ISMIL [22]	0.8600
Bag-based MIL [21]	0.8120
Multichannel and Multispatial Attention CNN [1]	0.8503
GCN-based approach [23]	0.8657
Task-agnostic SSL-based ViT [12]	0.8556
Patch-level CLA [11]	0.8468
Patch-level SA [11]	0.8402
TSOR	0.8843

SSL frameworks which were pretrained on a large database of histopathology images. In addition, no mechanism was incorporated to handle the stain variation. Thus, TSOR, pretrained using a stain-agnostic and task-specific SSL framework, performed consistently better compared to these methods.

Comparison was also made with histopathology-specific foundation models like CtransPath [32], PathoDuet [45], Kimianet [31], CHIEF [46], GigaPath [47], UNI2 [33], and Virchow [34], on the same three metrics. The classification head in Module 3 of our framework was first incorporated on top of the feature extractor part of the pretrained foundation models, and then fine-tuned on the PANDA dataset. The circular bar plots in Fig. 9 depict the results in both datasets. It is observed that our framework performs consistently well, compared to the others; thus, demonstrating its robustness.

b) *Grading*: The grading performance of TSOR was compared to that of the SOTA ISUP grading-specific methods [1], [2], [9], [10], [11], [12], [21], [22], [23] in terms of kappa, as shown in Table IV. Patch-level annotations were used in [2], [9], [12], and [11] to perform supervised training to extract the corresponding patch-level features. Since patch labels were inferred from WSIs [10], supervised patch-level training becomes noisy and is reflected in its ISUP grading. While task-agnostic SSL-based pretraining frameworks were employed in [22], [23], and [12] for patch-level feature extraction, the ImageNet pretrained models were used in [21] and [1]. None of these approaches focused on handling the issues that occur due to stain-variation in WSIs.

Our ISUP grading framework TSOR, based on task-specific and stain-agnostic SSL, was able to outperform these methods without using patch-level annotations. Furthermore, the ordinal regression factor during the fine-tuning of the pretrained model helped improve the kappa score. Kappa scores of 0.52 by [9], 0.50 by [2], and 0.87 by our proposed TSOR, on SICAP data, further demonstrate that TSOR is able to generalize better.

Further, the comparative study of TSOR with some SOTA generic MIL-based models for the task of ISUP grading is depicted in Table V. The results clearly demonstrate that though these task-agnostic MIL approaches perform relatively better in prostate cancer detection, they struggle to achieve desirable performance in ISUP grading. This highlights the

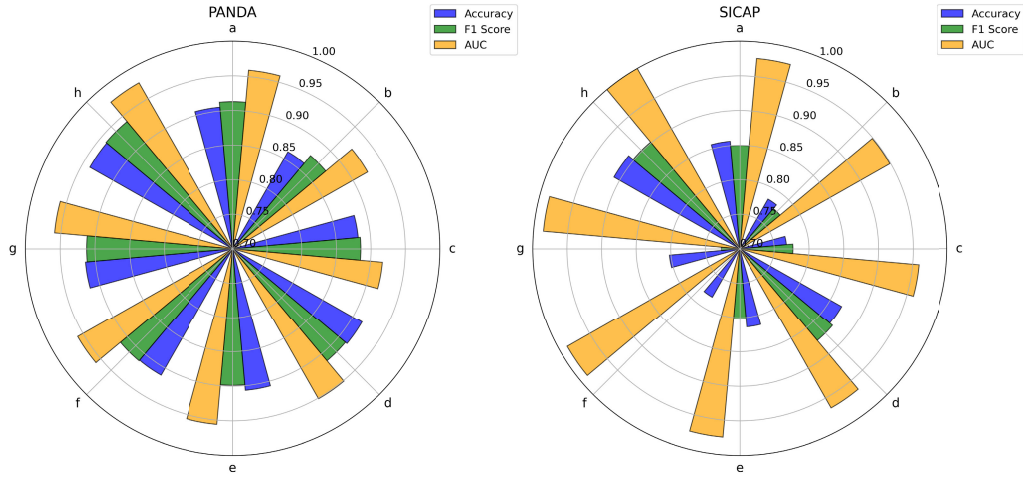


Fig. 9. Comparative performance of foundation models (a) CTransPath, (b) PathoDuet, (c) Kimianet, (d) CHIEF, (e) GigaPath, (f) UNI2, (g) Virchow, with our (h) TSOR, in terms of circular bar plots, for the detection of prostate cancer on PANDA and SICAP test sets.

TABLE V

COMPARATIVE STUDY WITH SOTA GENERIC MIL METHODS AND FOUNDATION MODELS, FOR ISUP GRADING ON TEST DATA, IN TERMS OF KAPPA

	Method	PANDA	SICAP
MIL-based model	CLAM [15]	0.6095	0.6563
	DSMIL [14]	0.6348	0.5787
	HIPT [18]	0.6579	0.6186
	CEMIL [19]	0.7063	0.6604
	2DMambaMIL [20]	0.6854	0.6572
	CTransPath [32]	0.8050	0.7441
Foundation model	PathoDuet [45]	0.6158	0.4977
	Kimianet [31]	0.7055	0.5705
	CHIEF [46]	0.8129	0.7138
	GigaPath [47]	0.8011	0.8309
	UNI2 [33]	0.8312	0.7193
	Virchow [34]	0.8368	0.8285
Proposed TSOR		0.8843	0.8736

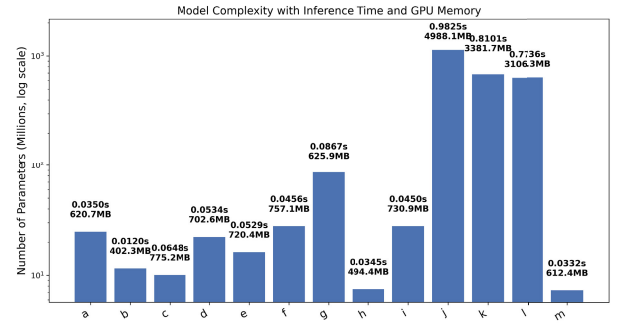


Fig. 10. Analysis of model complexity in terms of parameter count (bars) and inference time, peak GPU memory (annotated above each bar) of proposed (m) TSOR, against some SOTA MIL-based models: (a) CLAM, (b) DSMIL, (c) HIPT, (d) CEMIL, (e) 2DMambaMIL and foundation models: (f) CTransPath, (g) PathoDuet, (h) Kimianet, (i) CHIEF, (j) GigaPath, (k) UNI2, and (l) Virchow.

need for a task-specific framework tailored for prostate cancer grading.

The comparison of our TSOR with some SOTA foundation models (fine-tuned using PANDA data), for ISUP grading in terms of the kappa score, is also provided in Table V. It is observed that TSOR consistently outperforms the other models by a large margin. This shows that the use of a well-curated dataset for pretraining (compared to a large, diverse dataset of high-resolution histopathology images) allows a deep model to focus more on relevant information; thereby, resulting in notable improvement in performance. Also, the foundation models, pretrained on images of a fixed magnification, tend to underperform on images with different magnification.

The bar plot in Fig. 10 presents a comparative analysis of the model complexity of TSOR against other SOTA MIL frameworks and foundation models in terms of parameter count, inference time, and peak GPU memory usage during inference. Although MIL-based framework such as DSMIL exhibit slightly lower inference time and GPU memory usage than TSOR, and foundation model such as KimiaNet require marginally less GPU memory, these gains come at the cost of noticeably inferior agreement performance. As shown by

the kappa scores on PANDA and SICAP in Table V, TSOR consistently outperforms both conventional MIL-based models and foundation models by a substantial margin, while maintaining a low parameter count and competitive inference time and memory footprint. This demonstrates that TSOR achieves a favorable balance between model efficiency and grading performance, offering clear advantages over both SOTA MIL frameworks and foundation models.

3) *Explainability*: To provide confidence to pathologists, an automated diagnostic system should offer explainability regarding the interpretation of a given WSI of prostate cancer. The attention module of our TSOR helps identify the contribution of different patches to the final WSI grading, by assigning corresponding weights. Patches receiving greater weight should indicate regions of high diagnostic importance. Fig. 11 shows the patches of some of the annotated cancerous WSIs, that were weighted high. It is evident that the patches that are assigned a greater weightage by the model are actually relevant regions, as validated from the corresponding masks provided by expert pathologists. This shows that the

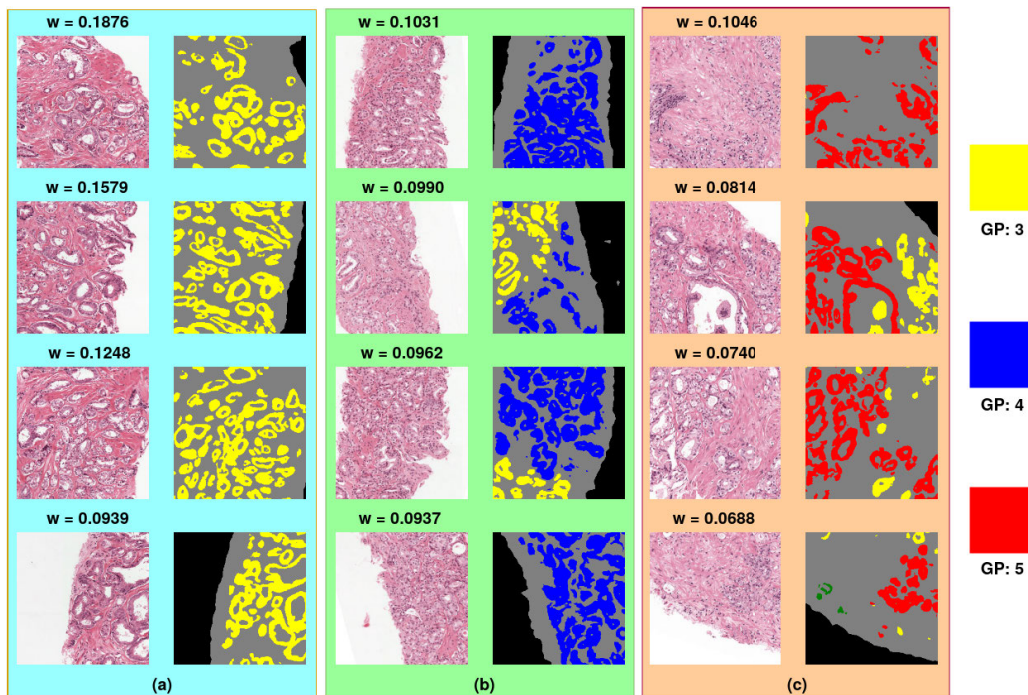


Fig. 11. Sample patches of high diagnostic importance, in terms of the weights [see (8)] predicted by TSOR, for three images from test data having ISUP grades (a) 1, (b) 3, and (c) 4. Each patch has its corresponding mask to its immediate right.

decision provided by TSOR is interpretable for the model to be considered reliable by human end users (here, pathologists).

IV. CONCLUSION AND DISCUSSION

An efficient SSL-based prostate cancer detection and ISUP grading framework was developed for the WSI of prostate biopsies, using only slide-level labels. Cancerous patterns in prostate cancer are highly heterogeneous and diverse, with stain variation between centers further affecting the generalizability of the models. In this context, it is essential to initially learn the patch-level features of GPs for effective WSI-level grading. Previous methods in the literature for prostate cancer detection or grading used supervised patch-level classifiers trained using expensive patch-level annotations [2], [9], [11], [12], ImageNet pretrained models with huge domain discrepancy [1], [15], [17], [21] or computationally expensive task-agnostic SSL frameworks [12], [14], [16], [22], [23] to extract patch-level features. In addition, none of these methods focused on handling the stain variation across centers.

In the first module of TSOR, a well-balanced patch-level dataset was created using pseudo-labeling within an efficient hybrid MIL framework based on slide-level labels only. The purpose of this intermediate dataset is to obtain a diverse and relatively balanced set of patches across GPs. This is done to ensure that the SSL model learns the stain-agnostic patch-level morphological features from all relevant patterns which are critical for effective WSI grading. Since SSL is label-agnostic and relies on intrinsic image-level transformations rather than explicit class labels, minor inaccuracies in the pseudo-labeling step do not propagate significant bias into the learned features.

Finally, the fine-tuning of the pretrained network was performed in terms of ordinal regression, in contrast to the

common approach of using one-hot encoded labels with cross-entropy loss. This helped TSOR to make fewer severe prediction errors (as observed from the confusion matrices); thereby, making it better suited for medical image grading. Also, the use of nonparametric classifiers like k NN [9], [10] for final grading has the inherent problems of sensitivity to noise, choice of k and high dimensionality. The consistent performance of TSOR across different datasets while outperforming the SOTA algorithms by considerable margin, as evidenced from the experimental results, indicated the effectiveness of our pretrained neural network-based method for ISUP grading. In contrast, the comparative study suggested a severe drop in the performance of the other methods when tested on external datasets for ISUP grading.

The proposed patch-level task-specific SSL pretraining also helped our framework to achieve good performance in prostate cancer detection, while remaining unaffected by the inherent class-imbalance problem among cancerous and noncancerous classes during detection. A high AUC value across datasets suggested that our framework is effective in identifying true positives while minimizing false positives. Thus, the TSOR framework achieved performance comparable to pathologists in both prostate cancer detection and grading tasks, using only intermediate resolution images that involve only slide-level labels readily available from clinical records. Moreover, the interpretability associated with the results justifies the trustworthiness of our framework.

Histopathology-specific foundation models were found to perform better in prostate cancer detection, compared to their grading. This shows that foundation models do not always perform well in grading tasks, where cancerous patterns can be heterogeneous and complex. This is perhaps because these

models, trained on diverse datasets, run the risk of over-generalization. Hence, they are unable to effectively capture many of the discriminating features required in ISUP grading. Moreover, these models are considerably heavyweight; thus, constraining their deployment in limited-resource environments. In contrast, TSOR is lightweight and based on a new stain-agnostic and task-specific SSL framework.

As prostate cancer is one of the leading causes of cancer-related deaths among men worldwide, the clinical relevance of this research is significant. The ISUP grading procedure by pathologists is time consuming and highly prone to intra- and inter-observer variability. Thus, the ability to provide an accurate and interpretable automated grading system can assist pathologists in making more consistent and timely diagnoses with high confidence; particularly in regions with limited access to specialized medical expertise. The high kappa score, achieved by our framework across different datasets, suggests a strong agreement between the predicted and actual labels, with the errors being mostly adjacent to the actual grade. The automated system, through the reduction in the workload of healthcare professionals and the minimization of diagnostic variability, has the potential to improve patient outcomes and enhance overall treatment efficiency.

The TSOR framework also has a strong potential to be adapted in other clinically relevant computational pathology tasks, such as biomarker prediction or prognostic modeling. For ISUP grading, the pseudo-labeled patches are relatively balanced across GGs. Similarly, for biomarker prediction or prognostic modeling, the SSL training dataset can be balanced across biomarker status or clinically defined outcome groups (e.g., low- vs high-risk cases) derived from slide-level labels. This strategy ensures that the SSL stage is exposed to comparable morphological diversity across clinically relevant groups, preventing representation bias toward dominant phenotypes. While the SSL-based pretraining stage of TSOR remains unchanged across tasks, the downstream prediction head and loss function need to be adapted according to the clinical endpoint. For biomarker prediction, the ISUP-specific ordinal head can be replaced with a binary or multilabel classification head trained using binary cross-entropy loss. For prognostic modeling, the classification head can be substituted with a risk prediction head optimized using survival-specific objectives such as the Cox partial likelihood.

Some of the limitations of the study point to the potential future scope. These include the availability of GPUs with considerable memory and the scarcity of publicly available annotated prostate cancer datasets. Although this method gives good performance by empirically selecting a constant number of patches from each WSI during training, it would be interesting to consider using other patch selection strategies with fewer WSIs for training. It would also be interesting to explore other SSL paradigms for patch-level pretraining. Aggregating patch-level features, using spatial-context-aware architectures like vision transformers (ViTs) and graph convolution networks (GCNs) would also be an interesting research direction. In addition, other explainability techniques can be explored in such MIL scenarios.

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